

Instructions are 8 bits : iiiiii

There are two types of instructions, based on the **value** of the **MSB** :

1- Memory instructions (MSB = 0) :

There are two types of memory instructions, based on the **value** of **the bit before the**

MSB :

- **Load** (to R register from RAM) [bit before MSB = 1]

- **Write** (to RAM register from R register) [bit before MSB = 0]

Either way, the syntax for the remaining bits is : adrs1adrs1 adrs2adrs2adrs2adrs2

where : -adrs1adrs1 is the 2 bit address of the relevant R register (00 to 11)

-adrs2adrs2adrs2adrs2 is the 4 bit address of the Relevant Register in the RAM (0000 to 1111)

2-Arithmetic Operations (MSB = 1) : these are performed by the ALU on two R register of choice, where the syntax for the remaining bits is : op op op adrs1adrs1 adrs2adrs2 where:

op op op are the 3 bits used to select the instruction of the ALU

adrs1adrs1 and adrs2adrs2 are the two bit addresses of the R register contents to be used in the 1st and 2nd ALU inputs respectively.

The result of ALU operations gets stored in the **second** R register (with address adrs2adrs2)

Some examples :

Instruction (Assembly)	Encoding
Load R1 1010	01 00 1010
Write R2 0110	00 01 0110
ADD R1 R2	1 110 00 01
SUB R3 R4	1 111 10 11
OR R1 R3	1 010 00 10
AND R2 R4	1 001 01 11
NOT R3	1 000 10 10 <i>*result gets stored in relevant register</i>